

In the following, we provide background for a nonlinear inverse problem relying on the Arrhenius relationship for thermally activated semi-conduction in minerals. The model follows the exponential form $\sigma(T) = \sigma_0 e^{-A/(kT)}$, where $\sigma(t)$ is the electrical conductivity at temperature T in degrees Kelvin, k is Boltzmann's constant of course, and A is the activation energy. This model is used to model the electrical conductivity data for the mineral olivine as a function of temperature. Olivine is a major constituent of Earth's mantle, and Dunite is an ultramafic plutonic rock that is composed almost exclusively of olivine.

We (a) describe the physical background, and (b) explain the geophysics/geology of the materials described, assuming only a very basic level of geologic background. We close each section with a summary.

1. Physical Background: Solid-State Conduction

In the context of electrodynamics, we generally treat materials as ohmic conductors where the current density is $\mathbf{J} = \sigma \mathbf{E}$. However, in minerals like olivine, σ is not a constant but a highly sensitive function of the microscopic state.

The Microscopic Mechanism

In high-temperature silicate minerals, electrical conduction is typically **thermally activated**. Since these minerals are insulators at low temperatures (wide band gaps), conduction relies on the movement of point defects rather than free electrons in a conduction band. The primary carriers are:

1. **Ionic conduction:** Transport of ions (e.g., Mg^{2+} , Fe^{2+} , or O^{2-}) through the crystal lattice via Vacancy or Interstitial mechanisms.
2. **Small polaron hopping:** The movement of an "extra" electron associated with a local lattice distortion, typically involving the valence exchange of iron ($Fe^{2+} \rightleftharpoons Fe^{3+}$).

Statistical Mechanics & The Arrhenius Form

The probability of a carrier overcoming a potential energy barrier A to move to an adjacent site is governed by the Boltzmann distribution. The jump frequency ν follows:

$$\nu = \nu_0 e^{-\frac{A}{kT}}$$

Where ΔG is the Gibbs free energy of activation ($G = H - TS$). In most geophysical applications, we focus on the activation enthalpy ($H = E + PV$), which simplifies to your activation energy term A at constant pressure.

To derive the conductivity, we look at the **Nernst-Einstein relation**, which links the diffusion coefficient D of the carriers to the electrical conductivity:

$$\sigma = \frac{nq^2D}{kT}$$

Where n is the charge carrier density and q is the charge. Since the diffusion coefficient D follows its own Arrhenius relationship $D = D_0 e^{-A/kT}$, the total conductivity takes the form:

$$\sigma(T) = \frac{\sigma'_0}{T} e^{-A/(kT)}$$

In practice, the $1/T$ pre-exponential dependence is often weak compared to the exponential term over the temperature ranges of interest (1000 K to 2000 K). Thus, it is mathematically collapsed into the constant

σ_0 you provided in your model:

$$\sigma(T) = \sigma_0 e^{-A/(kT)}$$

Summary of Physics

- **Carrier Dynamics:** Conduction in minerals is a stochastic process where charge carriers (ions or polarons) "hop" over energy barriers.
- **Temperature Sensitivity:** Because of the exponential dependence, a small change in temperature T leads to orders-of-magnitude changes in σ .
- **Activation Energy:** A represents the energy required to both create a defect and move it through the lattice.

2. Geophysics and Geology: Olivine and the Mantle

To understand why we use this specific model, we have to look at where these materials live and what they represent.

Olivine: The "Mantle Mineral"

The Earth is chemically layered. Beneath the thin crust lies the **mantle**, which extends to a depth of about 2,900 km. The upper mantle (down to 410 km) is dominated by the mineral **olivine** ($(\text{Mg, Fe})_2\text{SiO}_4$).

- It is a magnesium-iron silicate.
- It is typically olive-green in color (hence the name).
- In the mantle, it exists in a "mushy" solid state—it is solid, but over millions of years, it flows like a very viscous fluid.

Dunite: The Rock Perspective

While olivine is a mineral, **Dunite** is the rock type. If you have a rock that is more than 90% olivine, geologists call it Dunite.

- Dunite is an **ultramafic** rock, meaning it is very low in silica and high in magnesium and iron.
- In the field, you might find Dunite in "ophiolites" (pieces of the ocean floor and mantle thrust onto land) or as "xenoliths" (chunks of the mantle brought up by volcanic eruptions).

Why Electrical Conductivity Matters

Seismology tells us how fast waves travel, which gives us "stiffness" and "density." However, electrical conductivity tells us about **temperature** and **composition** (like water content).

1. **Thermal Probes:** Since σ is so sensitive to T , measuring the conductivity of the mantle via surface magnetic fields allows us to estimate the temperature of the Earth's interior without going there.
2. **The Role of Iron:** The "semi-conduction" mentioned in your model is largely due to the iron content in the olivine. Pure magnesium olivine (Forsterite) is a very poor conductor; the presence of iron facilitates the "hopping" of charges.

Summary of Geology

- **Composition:** Olivine is the most abundant mineral in the Earth's upper mantle; Dunite is the rock form of nearly pure olivine.
 - **Geophysical Proxy:** We use the Arrhenius relationship to map electrical conductivity (measured from the surface) into temperature (at depth).
 - **Environment:** These minerals operate under high pressure and temperature, where solid-state diffusion becomes the dominant form of electrical transport.
-